A FINITE UNION OF CONSTRUCTIBLE SUBSETS IS REFINED BY A FINITE CONSTRUCTIBLE STRATIFICATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of constructible subsets as finite unions of differences of retrocompact opens, and the constructibility of the pieces used in the family. Invoked inside the counted proof (source lines 5832-5834 and 5838-5840). Writes each piece as a difference of retrocompact opens and shows each member of the resulting family is constructible.

Source reference. Original lemma lemma-constructible: sources/stacks-topology/pinned-original-source.tex, source lines 2788–2788. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Suppose $X = T_1 \cup \ldots \cup T_n$ is written as a union of constructible subsets. There exists a finite stratification $X = \coprod X_i$ with each X_i constructible such that each T_k is a union of strata.*

Proof. By definition of constructible subsets, we can write each T_i as a finite union of $U \cap V^c$ with $U, V \subset X$ retrocompact open. Hence we may assume that $T_i = U_i \cap V_i^c$ with $U_i, V_i \subset X$ retrocompact open. Let S be the finite set of closed subsets of X consisting of $\emptyset, X, U_i^c, V_i^c$ and finite intersections of these. If $Z \in S$, then Z is constructible in X (Lemma lemma-constructible). Moreover, $Z \cap Z' \in S$ for all $Z, Z' \in S$. Define a partial ordering on S by inclusion. For $Z \in S$ set $X_Z = Z \setminus \bigcup_{Z' < Z,\ Z' \in S} Z'$ to get a stratification $X = \coprod_{Z \in S} X_Z$ satisfying the properties stated in the lemma. $\square$

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